

1. Lambda Function

Definition:

A lambda function is a small anonymous function used for short, one-line operations.

Syntax:

```
lambda arguments: expression
```

Example 1:

```
add = lambda a, b: a + b
print(add(3, 5))
```

Output:

```
8
```

Example 2:

```
check = lambda x: "Positive" if x > 0 else "Negative"
print(check(-4))
```

Output:

```
Negative
```

2. map() Function

Definition:

map() applies a function to every element in an iterable and returns a map object.

Syntax:

```
map(function, iterable)
```

Example 1:

```
nums = [1, 2, 3]
result = list(map(lambda x: x * 2, nums))
print(result)
```

Output:

```
[2, 4, 6]
```

Example 2:

```
nums = [1, 2, 3, 4]
result = list(map(lambda x: x*2 if x % 2 == 0 else x, nums))
print(result)
```

Output:

```
[1, 4, 3, 8]
```

3. filter() Function

Definition:

`filter()` selects elements from an iterable based on a condition.

Syntax:

```
filter(function, iterable)
```

Example 1:

```
nums = [1, 2, 3, 4, 5]
result = list(filter(lambda x: x > 2, nums))
print(result)
```

Output:

```
[3, 4, 5]
```

Example 2:

```
nums = [10, 15, 20, 25]
result = list(filter(lambda x: True if x % 2 == 0 else False, nums))
print(result)
```

Output:

```
[10, 20]
```

4. reduce() Function

Definition:

`reduce()` applies a function cumulatively to elements of an iterable and reduces it to a single value.

Syntax:

```
from functools import reduce

reduce(function, iterable)
```

Example 1:

```
from functools import reduce
nums = [1, 2, 3, 4]
result = reduce(lambda x, y: x + y, nums)
print(result)
```

Output:

```
10
```

Example 2:

```
from functools import reduce
nums = [5, 2, 8, 1]
result = reduce(lambda x, y: x if x > y else y, nums)
print(result)
```

Output:

```
8
```